

Assignment:- 2

Matrix Multiplication

1. Creation of text file

matrix.txt file will open in nano input file

Add your text in txt file -

For example -

A,0,0,1
A,0,1,2
A,1,0,3
A,1,1,4
B,0,0,5
B,0,1,6
B,1,0,7
B,1,1,8

Once the text is copy-pasted/ Added

Press ctrl + o

Then, Enter key

Lastly, Press ctrl + x

This will save the txt file and roll-back you to terminal

2. Create Mapper code file

After this command mapper_matrix.py will open in nano

Following is the code -

```
#!/usr/bin/env python3
import sys

# Dimensions (hardcoded or passed through env/config)
# A is m x n
# B is n x p
m = 2 # rows in A
n = 2 # cols in A / rows in B
p = 2 # cols in B

for line in sys.stdin:
    line = line.strip()
    if not line:
        continue
    matrix, i, j, value = line.split(",")
    i = int(i)
    j = int(j)
    value = float(value)
```

```

if matrix == "A":
for col in range(p):
    # Key: i,col ; Value: A,j,value
    print(f"{i},{col}\tA,{j},{value}")
elif matrix == "B":
for row in range(m):
    # Key: row,j ; Value: B,i,value
    print(f"{row},{j}\tB,{i},{value}")

```

In above code -

#!/usr/bin/env python3 is important to add.

Once the py code is complete

- Press ctrl + o
- Then, Enter key
- Lastly, Press ctrl + x

This will save the py file and roll-back you to terminal

3. Create Reducer code file

After this command reducer_matrix.py will open in nano

Following is the code -

```

#!/usr/bin/env python3
import sys
from collections import defaultdict

current_key = None
a_vals = defaultdict(float)
b_vals = defaultdict(float)

def emit_result(key, a_vals, b_vals):
    total = 0
    for k in a_vals:
        if k in b_vals:
            total += a_vals[k] * b_vals[k]
    print(f"{key}\t{total}")

for line in sys.stdin:
    line = line.strip()
    if not line:
        continue

    key, val = line.split("\t")
    if key != current_key and current_key is not None:
        emit_result(current_key, a_vals, b_vals)

```

```

a_vals.clear()
b_vals.clear()

current_key = key
tag, k, v = val.split(",")
k = int(k)
v = float(v)

if tag == "A":
    a_vals[k] = v
elif tag == "B":
    b_vals[k] = v

if current_key is not None:
    emit_result(current_key, a_vals, b_vals)

```

In above code -

#!/usr/bin/env python3 is important to add.

Once the py code is complete

- Press ctrl + o
- Then, Enter key
- Lastly, Press ctrl + x

This will save the py file and roll-back you to terminal

If your command went wrong

Re-correct it. Before running that command make sure you delete the output created using following command -

```
hduser@Ubuntu:~$ hadoop fs -rm -r /output_matrix
```

```

2025-04-14 18:00:47,700 WARN util.NativeCodeLoader: Unable to load native-hadoop library
for your platform... using builtin-java classes where applicable
Deleted /output

```

And then give your command -

```
hduser@Ubuntu:~$ hadoop jar /usr/local/hadoop/share/hadoop/tools/
lib/hadoop-streaming-3.3.4.jar -input /
mapreduce_matrix_multiplication/matrix.txt -output /output_matrix
-mapper /home/hduser/mapper_matrix.py -reducer /home/hduser/
reducer_matrix.py
```

Or else you will get following error -

```

2025-04-14 18:00:19,901 WARN util.NativeCodeLoader: Unable to load native-hadoop library
for your platform... using builtin-java classes where applicable
2025-04-14 18:00:20,774 INFO impl.MetricsConfig: Loaded properties from hadoop-
metrics2.properties
2025-04-14 18:00:21,121 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at
10 second(s).
2025-04-14 18:00:21,122 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2025-04-14 18:00:21,163 WARN impl.MetricsSystemImpl: JobTracker metrics system already
initialized!

```

2025-04-14 18:00:21,374 ERROR streaming.StreamJob: Error Launching job : Output directory
hdfs://localhost:54310/output **already exists**
Streaming Command Failed!